



Process Performance Management – Illuminating Design Issues through a Systematic Problem Analysis

Anne Cleven

Institute of Information Management,
University of St. Gallen
Müller-Friedbergstrasse 8
CH-9000 St. Gallen
+41 71 224 21 92

anne.cleven@unisg.ch

Robert Winter

Institute of Information Management,
University of St. Gallen
Müller-Friedbergstrasse 8
CH-9000 St. Gallen
+41 71 224 21 90

robert.winter@unisg.ch

Felix Wortmann

Institute of Information Management,
University of St. Gallen
Müller-Friedbergstrasse 8
CH-9000 St. Gallen
+41 71 224 37 88

felix.wortmann@unisg.ch

ABSTRACT

Business processes are the means by which organizations create value. Consequently, organizations need to continuously monitor and control their processes' performance so as to provide a consistent and predictable execution quality. A number of today's organizations, however, appear to encounter difficulties with measuring and improving their processes' performance. In this paper, we set out to identify the gap between how organizations currently approach process performance management (PPM) and what they are striving to realize in the future. The systematic gap analysis results in a set of design factors that are valuable in guiding future design efforts for useful and relevant PPM solutions.

Categories and Subject Descriptors

C.3 [Special-purpose and application-based systems]: Process control system, J.7 [Computers in other systems]: Process control

General Terms

Management, Measurement, Performance, Design

Keywords

Business Process Management, Process Performance Management, Problem Analysis, Empirical Research, Design Research

1. INTRODUCTION

Organizational efficiency, innovativeness and high productivity represent key requisites for organizations to secure their viability. In the light of the current economic crisis sustaining these capabilities takes on even greater significance. Providing an organization with the metrics and measures to survive in a fiercely competitive environment constitutes the core of the field of corporate performance management [9, 29]. Numerous research efforts from practitioners and researchers alike have brought about a multitude

of different approaches, the most famous being the Balanced Scorecard [18]. Only lately, however, have performance measurement and management concepts entered the field of business process management (BPM) so as to provide a detailed understanding of process performance issues and to foster an organization's flexibility, adaptability, and competitiveness [22]. It is thus hardly surprising that a number of surveys and research articles report on acute difficulties companies experience with a successful implementation [28, 34]. Both measurement and improvement of organizational performance on a process level appear to be highly challenging. In a 2009 international survey Wolf et al. reported that only 29% of the companies polled have always or at least in the majority of cases defined performance measures in place for evaluating the success of their major processes [35]. With 16% the share of organizations always or at least in the majority of cases using performance data to effectively manage their processes is even smaller [35].

While a number of surveys attest the *existence* of organizations' difficulties with implementing PPM there is a paucity of empirical research that explores *how* the problem is actually characterized. A precise understanding of the problem, however, is a critical prerequisite for the development of innovative artifacts to solve it. The problem analysis presented in this paper is part of a larger research project, which is following the design science research paradigm [16]. The project aims at developing artifacts supporting the alignment between PPM needs articulated by the business and adequate information technology (IT). Accordingly, the aim of this paper is to systematically analyze problems encountered with the development and accomplishment of PPM initiatives. In management science problems are commonly understood as a gap "between some existing situation and some desired situation" [25, p. 5]. Based on this comprehension problem definitions can be interpreted as the precise specification of such gaps. In order to define a problem, one thus compares "what is with what ought to be" [31, p. 968]. Hence, the research questions (RQ) that are driving the herein reported study are formulated as follows:

RQ1 – Which factors characterize current PPM approaches?

RQ2 – Which factors characterize desired PPM approaches?

RQ3 – What are the most popular development trends and what changes do they require?

We address these questions through a combined approach of survey and case study research. The survey we conducted is exploratory in nature and allows for deriving insights into characteristics

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee.

SAC'11, 21-MAR-2011, TaiChung, Taiwan

Copyright © 2011 ACM 978-1-4503-0113-8/11/03...\$10.00.

of both current and desired PPM approaches. It furthermore facilitates a detailed analysis of both the most popular movements between the two stages and the type of changes that have to be made for the transition. The subsequent case analyses serve two purposes: On the one hand they are employed as a means to validate the survey results. On the other hand, they provide a deeper understanding of the nature and complexity of PPM initiatives in organizations. The case interviews thus enable a more profound appreciation of the problem we address and the generation of additional suggestions for improvements and future work.

The remainder of this paper is structured as follows. In the subsequent section, we provide the conceptual background for our research by introducing the concepts of BPM and PPM. The third section introduces the research methodology. Section four presents the detailed problem analysis and the results that were derived. Based on the analysis we then propose how our results can serve as a basis for future research efforts aiming at the design of innovative PPM solutions (section 5). The paper concludes with a summary and a brief outlook on future research activities.

2. BACKGROUND

BPM as a holistic management concept prescribing that all organizational and management activity be based on the business's core processes has been around for more than one and a half decades now [2, 21]. The concept of BPM builds upon the notion of a business process as a series of cross-functional activities that need to be performed in order to collectively achieve a predefined goal [6, 15]. Core processes are frequently termed an organization's "strategic assets" [30] and describe end-to-end work that covers the whole value chain. The shift to a horizontal, process-oriented structure occurred as a result of increasing inefficiencies organizations encountered with a strictly vertical and function-oriented work structure. Starting with Frederick Taylor—who revolutionized work organization with task decomposition and job measurement—a number of concepts like statistic process control (1950s), Total Quality Management (TQM, 1980s), Six Sigma (1980s), and Business Process Reengineering (1990s) have further reinforced the trend towards process orientation [4]. Obtaining a comprehensive process orientation throughout the whole organization, however, is a non-trivial goal and only became practicable with the introduction of IT in the 1980s and 1990s [4].

Up to today a great plenty of organizations are still in the early steps of becoming consistently process-oriented as they are just about to analyze, truly understand and intentionally (re-)design their work practices. Since it requires allocating organizational as well as IT resources, defining policies and regulations, and thoroughly considering contextual environments and risk factors implementing BPM is a highly complex task [32]. In recent years both practitioners and academia have contributed immensely to the body of BPM knowledge providing a number of best practice frameworks as well as well-grounded insights into BPM as a research phenomenon. Much work has been carried out in the areas of business process modeling and design [5], managing information about processes [7], and process-aware information systems [27], to name but a few. One of the latest hot topics in the field of BPM is termed PPM. PPM is fuelled by an enormous practitioner interest. The concept accounts for the fact that once business processes are defined and in place, i.e., an organization has moved further in the BPM lifecycle it "will need to begin measuring processes and their outputs, and to continually refine their

designs" [7, p. 60]. The most elaborate definition of the notion PPM has so far been provided by practitioners. The *Association of Business Process Management Professionals* (ABPMP), one of the premier non-for-profit practitioner organizations in the field of BPM, defines the measurement of process performance as "the formal, planned monitoring of process execution and the tracing of results to determine the effectiveness and efficiency of the process" [1, p. 22]. The information gained in the measurement process is then "used to make decisions for improving or retiring existing processes and/or introducing new processes in order to meet the strategic objectives of the organization" [1, p. 22]. It becomes obvious from this definition that PPM consists of two major building blocks: Measurement and improvement of business process performance. Although the term PPM has as of yet not reached a broad popularity and the concept is still about to develop, there have been a number of prior efforts in both research and practice addressing either the measurement or improvement of business processes performance, or sometimes even both. Some of these approaches represent broad management concepts, like e.g., TQM or Six Sigma while others offer specific techniques for solving clearly delineated sub-problems of the whole challenge of PPM, like e.g., business activity monitoring, process mining, or simulation [20, 26]. A rather recently introduced concept, termed (business) process intelligence, builds on applying business intelligence (BI) techniques like data mining and statistical analysis for analyzing process data in order to uncover weaknesses and discover opportunities for improvement [12]. While BI has traditionally provided historical, retrospective analyses, in the context of process intelligence it is used to take the "pulse of the company" [14, p. 3] generating findings based on present-day transactional data that are analyzed in near real time. Drawing on the brief analysis of extant literature in the field of PPM it appears that the perception of what constitutes PPM and which approaches represent valuable components for its implementation are fairly undisputed. Nonetheless—as has been pointed out earlier in this paper—a great plenty of today's companies have not yet succeeded in establishing reasonable and integrated end-to-end PPM solutions [34, 35]. This perceived grievance motivated the subsequently described study.

3. RESEARCH METHODOLOGY

In order to address the above formulated research questions we employ a mixed-method approach integrating quantitative (survey) and qualitative (case study) research. It has frequently been argued that the use of different research methods will produce more copious and more reliable results [23]. Surveys and case studies are thus not viewed as competing sources of evidence, but as complementary methods that are incomplete without the other. For investigating the problem of developing and maintaining a successful PPM approach we first conducted an exploratory questionnaire-based survey that allowed us to achieve preliminary insights into current and desired approaches. Moreover, we analyzed the movements between the as-is and the to-be situations and thereby illuminated the biggest change issues. In order to both validate the survey results and facilitate a deeper understanding of the problem as well as a better appreciation of its nature and complexity, we subsequently accomplished three case studies with organizations representing three of the beforehand identified 'problem classes'. The subsequent sections describe in detail how data was collected and analyzed.

3.1 Survey

Survey research has a strong tradition in the Information Systems (IS) discipline [19]. It is frequently applied in order to gather knowledge on typical characteristics of larger populations in new and not yet well-known situations [24]. “By studying a representative sample of organizations, the survey approach seeks to discover relationships that are common across organizations and hence to provide generalizable statements about the object of study” [11]. Exploratory survey research is used during the early phases of research into a phenomenon, when the intent is to acquisition preliminary insight [10]. Commonly an exploratory survey is not based on an existing model, as the concepts of interest are yet to be identified and better understood. It instead helps to establish the concepts and discover new facets of the subject under research.

For developing the survey instrument an initial list of measurement items was prepared through a broad review of both the academic and professional literature on PPM. As has been shown above, PPM represents a multidisciplinary field of study that builds on a variety of concepts, methods and techniques from other research areas. The three main areas have been identified to be BPM, BI and performance management [4]. The questionnaire thus included items from each of these areas. The first version of the questionnaire was pretested with regard to wording, coherency, and ease of interpretation by three academics with adequate expertise. The instrument was revised according to their feedback. Before finally being distributed the questionnaire was again pretested by three practitioners with regard to consistency, understandability, and adequacy of item sequence. The described proceeding resulted in a final questionnaire covering 21 items that relate to the research phenomenon PPM. The questionnaire was distributed at a practitioner event on Business Intelligence (BI) and Business Analytics held in October 2009. The participants were specialists and executives working on both the IT and the business side, thus having the required knowledge and information to answer the questions [10]. In order to minimize the potential of semantic misunderstandings the questionnaire was introduced to the participants including explanations on the purpose and the basic concepts that are addressed. The questionnaire was designed to assess both the current and the desired state of PPM in the participating organizations. Respondents were asked to indicate the degrees of realization for each item using a five-tiered Likert scale. A total of 49 questionnaires were returned. If a data set was incomplete, i.e., if one or more than one of the 21 items was missing, the questionnaire was disregarded. On the basis of this criterion, 42 questionnaires were selected for further analysis. The surveyed organizations are primarily large and medium-sized companies from the German-speaking countries. 60% have more than 1000 employees and another 22% have more than 100 employees. The sectors mainly represented were professional services (40%), banking, finance and insurance (29%), high tech (11%) manufacturing and consumer goods (7%), media and telecommunication (5%), and others (8%). For analyzing the data we applied exploratory factor analysis (EFA) and cluster analysis. EFA serves the purpose of developing a deeper understanding of a specific problem situation and the identification of a number of important and mutually independent factors from a multiplicity of contingent variables [8]. The question of whether there are common situations which feature the same characteristics can be tackled using cluster analysis. A cluster analysis serves the pur-

pose of partitioning a set of observations into subsets that are homogeneous within and heterogeneous amongst them [17].

Two EFA were performed independently for as-is and to-be situations each based on the resp. data set of 21 items. The measure of sampling adequacy (MSA, “Kaiser-Meyer-Olkin criterion”) for the as-is (to-be) data set is 0.777 (0.767). MSA represents an indicator for the extent to which the input variables belong together and provides information on whether a factor analysis can reasonably be performed or not. Kaiser and Rice appraise a value of 0.7 and above as “reasonable”, i.e., the data set is considered appropriate for applying EFA [33]. For the as-is (to-be) situation five (four) factors that jointly explain about 75.6% (71.9%) of the total variance were extracted by means of principal component analysis. The resulting component matrices were rotated using the Varimax method with Kaiser normalization in order to improve the interpretability of the items’ assignment to the factors [17]. In order to identify organizations with similar as-is resp. to-be approaches to PPM, cluster analysis is used. The cluster analysis is based on factor scores that are calculated using the regression method. The Ward fusion algorithm and the squared Euclidean distance are applied for clustering, as this combination finds very good partitions resulting in an appropriate number of clusters and similar number of observations in each cluster [13]. On the basis of the dendrogram, i.e., the graphical representation of the fusion process and the cluster sizes the final numbers of clusters for both situations were defined [13]. The following section briefly outlines the conduct of the case studies.

3.2 Case Studies

Case studies and especially semi-structured interviews are considered highly useful for exploring areas in which research is in an early stage [3]. They allow the researcher to study a phenomenon directly as it is perceived by the subjects involved, learn about the state of practice, and understand real-world processes and decision making. Other than survey research case studies allow for answering ‘how’ and ‘why’ questions [3]. In order to better understand the how and why of PPM approaches case studies with three organizations were accomplished. The companies were selected so as to gain a deeper understanding of three distinct ‘problem classes’ that were identified through the survey. The attribution to a problem class was based on a self-assessment of the organizations. Data was collected via semi-structured face-to-face interviews with process and performance analysts, quality managers, and BI specialists. Findings presented below aggregate results from interview analysis per organization. Due to company request the cases are presented anonymously.

4. RESEARCH FINDINGS

4.1 Current Approaches to PPM

By means of the EFA we were able to identify five distinct factors that effectively characterize current PPM approaches in the surveyed organizations. The first factor, which we subsequently refer to as *Broad, plan-do-check-act (PDCA)-based use of PPM measures*, combines seven items. All of the items are metric-related and shed light on a) which key performance indicators (KPIs) are used for PPM and b) which method is applied to conduct PPM, namely the PDCA cycle. The second factor, termed *BPM maturity*, consolidates five items that indicate a strong process orientation as a central organizational paradigm. Another four items were found to load high on the third factor, which we named *PPM*

process diffusion. It reveals a high degree of KPI usage for the core business processes (covering procurement, production and sales) on the one hand and the importance of non-financial measures on the other. Factor four, in the following referred to as *BI-enabled, integrated PPM*, groups three items that together strengthen the need for well-defined and well-coordinated processes in both enterprise performance management and BI. The fifth and last factor, *High quality information base*, combines two items that point to the need of an integrated high quality data base for a consistent PPM approach. Building on these factors we conducted a cluster analysis so as to classify similar current approaches to PPM. We derived the following four clusters.

The first cluster is characterized by a poor efficacy with respect to four out of the five identified factors. Only PPM process diffusion, i.e., the degree of KPI usage in core business processes is at an above average level. The approach is in particular lacking a high quality information base. Neither from a BI perspective nor from a BPM perspective are fundamental concepts in place that could be leveraged as a basis for PPM. Due to low degree of consistency and intentionality we termed this approach *Isolated initiatives approach*. The second cluster shows a strong performance with respect to BPM maturity and the quality of the information base. Both a profound process orientation and consistently high information quality form a solid basis for PPM. Nevertheless, all three directly PPM related factors—Broad, PDCA-based use of PPM measures, PPM process diffusion, and BI-enabled, integrated PPM—show a low degree of implementation. The approach specifically lacks well-defined, well-coordinated processes and concepts in performance management as well as in BI. Although this approach offers a promising basis to establish PPM the fundamental components are not aligned, which let us call it *Fragmented approach*.

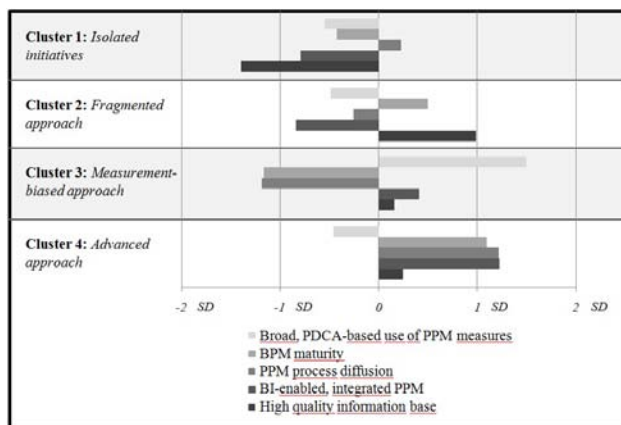


Figure 1: Arithmetic Means of Factors per as-is Cluster

The approach characterized through the third cluster puts a very strong focus on measurement. Organizations implementing this approach show an oddly high use of KPIs, which—considering the performance of the other factors—appears very ambitious. Although their BI landscape is integrated and rather sophisticated, essential BPM concepts that form the basis for PPM are not in place and thus hamper a comprehensive and successful implementation. Because of its very strong concentration on KPI measurement we call this cluster the *Measurement-biased approach*. The fourth cluster shows a high performance across four out of the five factors. The only weak factor is factor one. It is thus plausible

to call this approach the *Advanced approach*. It is characterized through a high BPM maturity and a well-established BI organization providing an adequate information base for PPM. Organizations using this approach apply performance measurement to a significant number of their business processes building upon a consistent set of limited metrics. For an easier interpretation and better comparison of the different approaches we standardized the cluster means. Figure 1 depicts how many standard deviations a cluster value is above or below the overall factor mean (across all cluster values).

4.2 Desired Approaches to PPM

The EFA we conducted on the data set describing the desired or to-be approaches resulted in four factors, which can be interpreted as design factors for developing these approaches. The first factor combines ten items that collectively describe the extent and sophistication of process performance measurement. We thus called it *Scope and quality of KPI measurement*. The second factor, which we subsequently refer to as *PPM coverage of core processes*, groups together the four items that describe the processes PPM is applied for and the need for non-financial measures. Four items load high on the factor subsequently called *BPM maturity*. The last factor, called *Integrated, methodologically sound PPM*, consolidates three factors that point to a strong methodology as regards performance management issues. The subsequently accomplished cluster analysis again led to four distinct approaches, which we describe in the following.

The first cluster represents approaches that put an emphasis on a broad set of KPIs and high quality decision making. Organizations striving to establish the approach subsequently called *Measurement-focused approach* do not focus on achieving a high BPM maturity. They rather want to apply a considerable number of metrics for measuring the performance of their core processes and—in so doing—rely on a high quality integrated data base. Not emphasizing a methodologically sound and strategically anchored PPM initiative these organizations aim at using PPM in an ad-hoc manner and not necessarily systematically aligned with corporate performance management issues. The *BPM-biased approach* represents a strong antagonism to the measurement-focused approach. Companies striving to launch this approach will put a strong emphasis on consistently documented processes that are transparent across functional and system borders. They neither want to attempt instituting PPM as an integral part of their organizations' overall performance management approaches nor favor an ad hoc or on demand measurement. Quite the contrary: Measurement is basically advent in this concept characterizing it as an approach rather meant to support the early phase of BPM adoption. The third cluster advocates an elementary PPM approach that is aligned with the organization's strategic performance initiatives. Clear responsibilities for processes as well as defined measurement practices play an important role. However, organizations striving to implement this approach plan to apply PPM only to a constrained number of processes, giving this concept the name *Selective approach*. The focus lies on a rigorous enterprise performance management with a strong measurement basis. Both non-financial KPIs and the measurement of organizational performance on a process level appear to have a supplementary character. The *full-scale approach* is characterized by a sound process performance measurement and management. Organizations striving to launch this approach intend to use a manageable number of performance metrics primarily for their core

processes. A broad and consistent process orientation is perceived essential and the strong conjunction of corporate and process performance management approaches points to the fact that these companies treat their processes as strategic assets. Again, we standardized the cluster means: Figure 2 depicts how many standard deviations a cluster value is above or below the overall factor mean (across all cluster values).

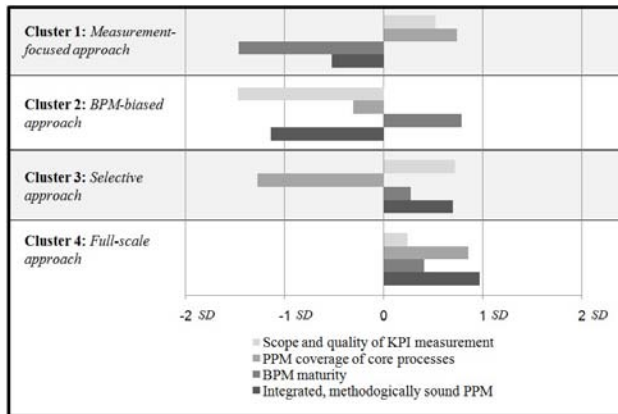


Figure 2: Arithmetic Means of Factors per to-be Cluster

While analyzing both the existing and the desired PPM approaches is an attractive research goal their mere description offers only limited use for gaining knowledge and achieving evidence on where improvement is required after all. The more intriguing questions thus are a) What are the most popular development trends between the clusters?, and b) What are the most important changes for the transition from a certain current to a more desired state? In order to answer these questions we analyzed the movement for each organization that participated in the survey and aggregated the findings. The next chapter describes the results.

4.3 Development Trends

The movements between the current approaches to PPM on the left-hand side and the desired approaches on the right-hand side are visualized in Figure 3.

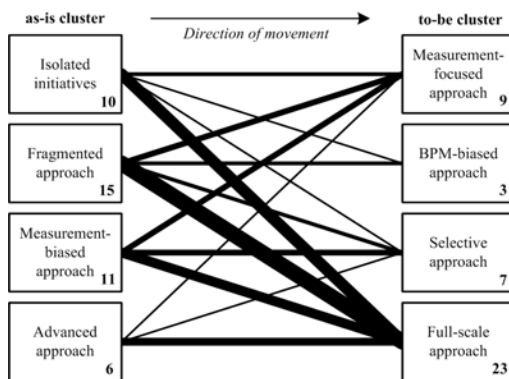


Figure 3: Development Trends

The number in the lower right corner of each of the squares indicates the population of each cluster. The thickness of the lines is proportionate to the number of movements. It is obvious—albeit not surprising—that most of the organizations are striving to prospectively realize the full-scale approach. On the contrary, a noticeably small number of organizations aim at implementing the BPM-biased approach. It is moreover interesting that the dis-

position of organizations across the different as-is approaches is comparatively homogenous while there is a clear preference structure discernible across the to-be approaches. Thus, although commencing from quite dissimilar starting positions organizations appear to largely agree on what is the most desirable approach. Due to the differing conditions, however, the ways to reach this position, i.e., to implement the more desired approach vary significantly.

In order to shed light on what are the most important changes to be made in order to obtain the intended PPM approaches we deeper analyzed the most interesting movements, which are the ones consisting of at least three movers. To this end we calculated the weighted adjacency matrix for the migration graph (cf. Figure 3) based on the items that are grouped by the to-be factors (cf. Table 1). The values represent the arithmetic means of the differences between the item values to-be and as-is across all movers per path. This allows us to identify the areas that require the most change when an enterprise intends to reach a certain to-be PPM approach. We chose to make the adjacency analysis on the item-level as opposed to on the factor level, because the aggregation of items into bunched concepts only allows for limited interpretability.

One of the major paths and the first movement described in the adjacency matrix is the shift from the isolated initiatives approach towards a full-scale PPM implementation. Organizations running isolated PPM initiatives are especially characterized by the lack of a high quality information base. Moreover, neither from a BI nor from a BPM perspective have these organizations established concepts in place. The more detailed analysis reveals that in order to realize an effectual PPM concept these companies explicitly focus on a) providing an adequate IT infrastructure and b) gaining a deeper knowledge about their work by consistently documenting their processes. Besides, in order to bundle thus far isolated initiatives organizations consider integrating their PPM ambitions with an enterprise BSC and/or their enterprise performance management program. Organizations starting off from the fragmented approach strive for either the measurement-focused or the full-scale approach. As described above the measurement-focused approach emphasizes the use of a broad set of KPIs for high quality decision making regarding the improvement of core processes—assumedly often in an on-demand setting based on real-time data. While organizations using the fragmented approach show a profound process orientation and possess a high quality information base the transition to the measurement-focused approach requires the development of a set of crucial process performance metrics like process quality, cycle times as well as resource and capacity utilization. Moreover, the change demands a stronger concentration on the measurement of core processes in particular on the basis of non-financial metrics. A shift from the fragmented to the full-scale approach on the contrary, first and foremost necessitates a more organized and well-planned method. In order to overcome the parallelism of a strong process orientation and an underutilized IT infrastructure, organizations rigorously implement the PDCA-cycle and align their process performance initiatives with the enterprise performance management. Bridging the business-IT-divide is further enhanced by making process flows transparent across functional borders for the whole workforce. The measurement-biased approach—as the name betrays—is characterized by a strong overestimation of KPI use and measurement. Consequently, each convergence towards

Table 1. Adjacency Matrix

as-is cluster		Isolated initiatives		Fragmented			Measurement-biased				Advanced	
				delta			delta				delta	
to-be factors Items	to-be cluster	as-is cluster means	Full-Scale	as-is cluster means	Measure-ment-focused	Full-Scale	as-is cluster means	Measure-ment-focused	Selective	Full-Scale	as-is cluster means	Full-Scale
Scope and quality of KPI measurement												
Adherence to schedules is measured for processes.		2.2	1.7	2.6	1.7	1.3	3.5	1.0	1.0	0.8	3.0	1.3
Quality is measured for processes.		1.6	2.2	2.2	2.7	1.4	3.0	0.7	1.3	1.4	2.7	1.5
Process cycle times are measured.		1.8	1.7	2.4	2.0	1.5	2.8	1.3	1.3	1.2	3.4	0.8
Process costs are measured.		2.1	1.8	2.3	2.0	1.3	2.8	0.7	1.3	1.8	2.9	1.3
Capacity utilization is measured for processes.		2.2	1.3	2.8	2.0	1.1	2.8	1.3	1.3	1.2	3.3	1.0
Process resource utilization is measured.		2.2	1.7	2.5	2.0	1.6	3.0	0.7	1.0	1.4	3.1	1.3
Data quality is consistently high.		1.7	2.7	3.4	0.7	1.1	3.1	1.7	1.0	1.4	2.7	1.5
A central integrated data base is in place (e.g., an Enterprise DWH).		1.6	2.3	3.8	1.0	1.0	3.1	1.3	1.3	0.8	3.7	0.8
Processes have defined process officers.		2.9	1.2	3.4	1.0	1.0	3.8	1.3	1.3	1.4	4.0	0.8
Defined BI governance responsibilities and processes are in place.		1.6	2.0	2.3	2.0	1.8	2.8	1.0	1.3	1.4	3.9	0.5
PPM coverage of core processes												
PPM is deployed for production processes.		2.1	1.0	2.2	1.7	1.6	2.5	0.3	0.0	1.0	4.0	0.8
PPM is deployed for sales processes.		2.4	1.0	2.8	1.0	1.1	2.0	0.7	0.0	1.4	3.4	1.3
PPM is deployed for purchasing processes.		2.0	1.2	1.9	1.7	1.3	1.6	0.7	0.0	1.8	2.9	1.3
PPM also covers non-financial measures.		2.4	1.0	1.8	1.7	0.9	2.1	1.0	0.3	0.6	3.1	-0.3
BPM maturity												
Process orientation is a central paradigm.		2.9	1.5	3.1	1.0	1.5	3.1	1.0	0.7	1.6	3.6	0.8
Process flows are consistent and transparent beyond functional borders.		2.4	1.3	2.9	0.7	1.8	2.3	1.7	1.7	2.4	3.4	1.0
Processes are consistently documented and/or modelled.		2.2	2.2	2.9	1.3	1.5	2.6	1.7	2.0	2.4	3.6	1.0
Process flows are consistent and transparent beyond system borders.		2.3	1.5	2.8	1.0	1.6	2.1	1.0	1.3	3.0	3.1	1.3
Integrated, methodologically sound PPM												
PPM is part of the enterprise-wide Balanced Score Card (BSC).		1.7	2.0	1.4	0.3	1.5	2.5	0.0	0.3	1.0	3.3	1.3
PPM is part of the Corporate Performance Management (CPM).		1.7	2.0	1.8	0.0	2.0	2.4	0.0	0.3	1.0	3.4	1.0
The plan-do-check-act (PDCA) cycle is applied for PPM.		1.4	1.7	2.0	0.0	2.1	2.7	0.0	0.3	0.8	3.3	1.3

another approach—be it measurement-focused, selective, or full-scale—requires establishing a more sophisticated process orientation. In any case the measurement-biased organization must consistently design and model its processes and make them transparent beyond both functional and system borders so as to enable a comprehensive and successful PPM implementation. The last movement we analyzed in more detail leads from the advanced to the full-scale approach. Due to the fact that the advanced approach already shows a comparatively high maturity only minor adoptions are required. The analysis reveals that an improvement of the overall data quality becomes necessary. As the measurement level of production processes is already very high, the focus will be shifted to sales and purchasing processes. An increased use of the BSC and the PDCA-cycle further fosters the holism of the approach. As has been mentioned in section three the survey we conducted was exploratory in nature and as such not based on well-known and empirically proven relationships. In order to validate our findings we thus complemented our analysis with three case studies. The selection of the cases was based on a self-assessment of the organizations when confronted with the survey results. Subsequently the case study results are presented in detail. All three organizations are implementing a full-scale PPM approach, however, each starting off from a different as-is position.

4.4 Validation of Results through Case Study Findings

4.4.1 Case 1 – Isolated initiatives towards full-scale

Company A is a globally operating insurance company. Due to a number of acquisitions their current IT landscape is highly heterogeneous and fragmented. Their movement towards a full-scale PPM approach was initiated by the parent company, which prescribed that all subsidiaries have an approach in place until 2012. While the persons in charge of the PPM initiative at the subsidiary we worked with have a strong ambition themselves to realize a holistic PPM concept, there are a number of obstacles to resolve on the way towards a successful implementation. The interviews revealed that on the one hand a perceived low market pressure and an entrenched functional work habit among the workforce hamper the establishment of a consistent process orientation and

process culture. On the other hand after an increased acquisition activity the responsibility for the IT landscape is broken up between two detached teams that show little willingness to work together. Hence, process-oriented thought and work patterns and enthusiasm for communicating and pursuing a shared goal represent additional setting levers on the way to an efficient PPM initiative.

4.4.2 Case 2 – Fragmented towards full-scale

Company B is a global player in the construction industry. As a manufacturing company employing a Lean Six Sigma approach for their production processes they have a sound process understanding and culture. Their IT landscape is extraordinarily well integrated. The company is managed through a sophisticated CPM approach. However, major difficulties in establishing the intended comprehensive PPM approach on the one hand lie in identifying the appropriate metrics and KPIs to purposefully measure process performance and on the other hand in building the link between the black-box financial performance management and a white-box approach measuring performance on a process level. Moreover, the strong process orientation found in the production facilities is not equally lived in the sales and research & development departments requiring additional process missionary work. The way to a full-scale PPM approach thus includes the definition of value drives trees and a fostering of process thinking throughout the whole company.

4.4.3 Case 3 – Measurement-biased towards full-scale

Organization C is one of the largest telecommunication companies in Switzerland. Its strong competitive position makes market pressure a minor concern. The main focus of management lies on new product development and sales figures. The current PPM approach represents a pilot project implemented within the boundaries of one department. The availability of an integrated IT infrastructure and a sophisticated PPM tool make nearly any kind of process performance measurement feasible. Dashboards have been developed allowing for not only a retrospective, but a real-time performance assessment. However, nearly no improvement initiatives are launched on the basis of the copious measurement results. The workforce currently shows a strong resistance to the

measurement initiatives for fear of complete transparency of the individual people performance. Moreover, the humble management valuation of the benefits attainable through an integrated full-scale PPM approach currently makes the pilot department a lone fighter. Consequently, in this case a lot of educational and convictional work regarding the benefits of PPM needs to be carried out. The subsequent chapter summarizes the findings and gives a brief outlook on future research.

5. CONCLUSION AND OUTLOOK

This paper provides a systematic analysis of the major issues and challenges related to PPM in organizations as identified through an exploratory survey and perceived by experts in the field. In order to illuminate design issues in the field we applied a rigorous approach, using both quantitative and qualitative research methods. By means of an exploratory survey we derived a set of four design factors that represent a relevant prerequisite for developing suitable and well-fitting PPM solutions. The subsequent use of case studies followed two purposes: On the one hand it allowed us to validate the identified approaches and the development trends. On the other hand it revealed additional requirements for the development of artifacts to support PPM initiatives. Future research efforts will use the herein gained knowledge to inform the design of situational PPM artifacts.

6. REFERENCES

- [1] ABPMP *Guide to the Business Process Management Common Body of Knowledge*. <http://www.abpmp.org>, 2009.
- [2] Armistead, C., Pritchard, J.-P., Machin, S. Strategic Business Process Management for Organizational Effectiveness. *Long Range Plan.*, 32, 1 (1999), 96-106.
- [3] Benbasat, I., Goldstein, D.K., Mead, M. The Case Research Strategy in Studies of Information Systems. *Manag. Inf. Syst. Quart.*, 11, 3 (1987), 369-386.
- [4] Bonham, S.S. *Actionable Strategies Through Integrated Performance, Process, Project, and Risk Management*. Artech House, 2008.
- [5] Damij, N. Business Process Modelling Using Diagrammatic and Tabular Techniques. *Bus. Process Manag. J.*, 13, 1 (2007), 70-90.
- [6] Davenport, T.H. *Process Innovation. Reengineering Work through Information Technology*. Harvard Business Press, Boston, MA, 1993.
- [7] Davenport, T.H., Beers, M.C. Managing Information about Processes. *J. Manag. Inf. Syst.*, 12, 1 (1995), 57-80.
- [8] de Winter, J.C.F., Dodou, D., Wieringa, P.A. Exploratory Factor Analysis With Small Sample Sizes. *Multivar. Behav. Res.*, 44, (2009), 147-181.
- [9] Folan, P., Browne, J. A Review of Performance Measurement: Towards Performance Management. *Comput. Ind.*, 56, (2005), 663-680.
- [10] Forza, C. Survey Research in Operations Management: A Process-based Perspective. *Int. J. Operations & Prod. Manag.*, 22, 2 (2002), 152-194.
- [11] Gable, G.G. Integrating Case Study and Survey Research Methods: An Example in Information Systems. *Eur. J. Inf. Syst.*, 3, 2 (1994), 112-126.
- [12] Grigori, D., Casati, F., Castellanos, M., Dayal, U., Sayal, M., Shan, M.-C. Business Process Intelligence. *Comp. Ind.*, 53, 3 (2004), 321-343.
- [13] Hair, J.F.J., Black, W.C., Babin, B.J., Anderson, R.E., Tatham, R.L. *Multivariate Data Analysis*. Pearson Prentice Hall, Upper Saddle River, New Jersey, 2006.
- [14] Hall, C. Business Process Intelligence. *BPTrends*, 2, 6 (2004), 1-11.
- [15] Hammer, M. Reengineering Work: Don't Automate, Obliterate. *Harv. Bus. Rev.*, 68, 4 (1990), 104-112.
- [16] Hevner, A.R. A Three Cycle View of Design Science Research. *Scand. J. Inf. Syst.*, 19, 2 (2007), 87-92.
- [17] Kaiser, H.F. The Varimax Criterion for Analytic Rotation in Factor Analysis. *Psychometrika*, 23, 3 (1958), 187-200.
- [18] Kaplan, R.S., Norton, D.R. The Balanced Scorecard: Measures That Drive Performance. *Harv. Bus. Rev.*, 83, 7/8 (2005), 172-180.
- [19] King, W.R., He, J. External Validity in IS Survey Research. *Commun. Assoc. Inf. Syst.*, 16, (2005), 880-894.
- [20] Kueng, P. Process Performance Measurement System: A Tool to Support Process-Based Organizations. *Total Qual. Manag.*, 11, 1 (2000), 67-85.
- [21] Lee, R.G., Dale, B.G. Business Process Management – A Review and Evaluation. *Bus. Process Manag. J.*, 4, 3 (1998), 214-225.
- [22] Martin, F. A Performance Technologist's Approach to Process Performance Improvement. *Perform. Improv.*, 47, 2 (2008), 30-40.
- [23] Mingers, J. Combining IS Research Methods: Towards a Pluralist Methodology. *Inf. Syst. Res.*, 12, 3 (2001), 240-259.
- [24] Palvia, P., Leary, D., Mao, E., Midha, V., Pinjani, P., Salam, A.F. Research Methodologies in MIS: An Update. *Commun. Assoc. Inf. Syst.*, 14, (2004), 526-542.
- [25] Pounds, W.F. The Process of Problem Finding. *Ind. Manag. Rev.*, 11, 1 (1969), 1-19.
- [26] Powell, S.G., Schwaninger, M., Trimble, C. Measurement and Control of Business Processes. *Syst. Dyn. Rev.*, 17, 1 (2001), 63-91.
- [27] Reijers, H.A. Implementing BPM Systems: The Role of Process Orientation. *Bus. Process Manag. J.*, 12, 4 (2006), 389-409.
- [28] Robson, I. From Process Measurement to Performance Improvement. *Bus. Process Manag. J.*, 10, 5 (2004), 510-521.
- [29] Saltmarsh, D., Ireland, J.M., McGregor, A. The Performance Framework: A Systems Approach to Understanding Performance Management. *Public Admin. Dev.*, 23, 5 (2003), 445-456.
- [30] Smart, P.A., Maddern, H., Maull, R.S. Understanding Business Process Management: Implications for Theory and Practice. *Br. J. Manag.*, 20, 4 (2009), 491-507.
- [31] Smith, G.F. Defining Managerial Problems: A Framework for Prescriptive Theorizing. *Manag. Sci.*, 35, 8 (1989), 963-981.
- [32] Smith, H., Fingar, P. *Business Process Management – The Third Wave*. Meghan-Kiffer Press, Tampa, FL, 2003.
- [33] Stewart, D.W. The Application and Misapplication of Factor Analysis in Marketing Research. *J. Marketing Res.*, 18, 1 (1981), 51-62.
- [34] Vergidis, K., Turner, C.J., Tiwari, A. Business Process Perspectives: Theoretical Developments vs. Real-world Practice. *Int. J. Prod. Econ.*, 114, 1 (2008), 91-104.
- [35] Wolf, C., Harmon, P. The State of Business Process Management 2010. *BPTrends*, (2010), 1-52.